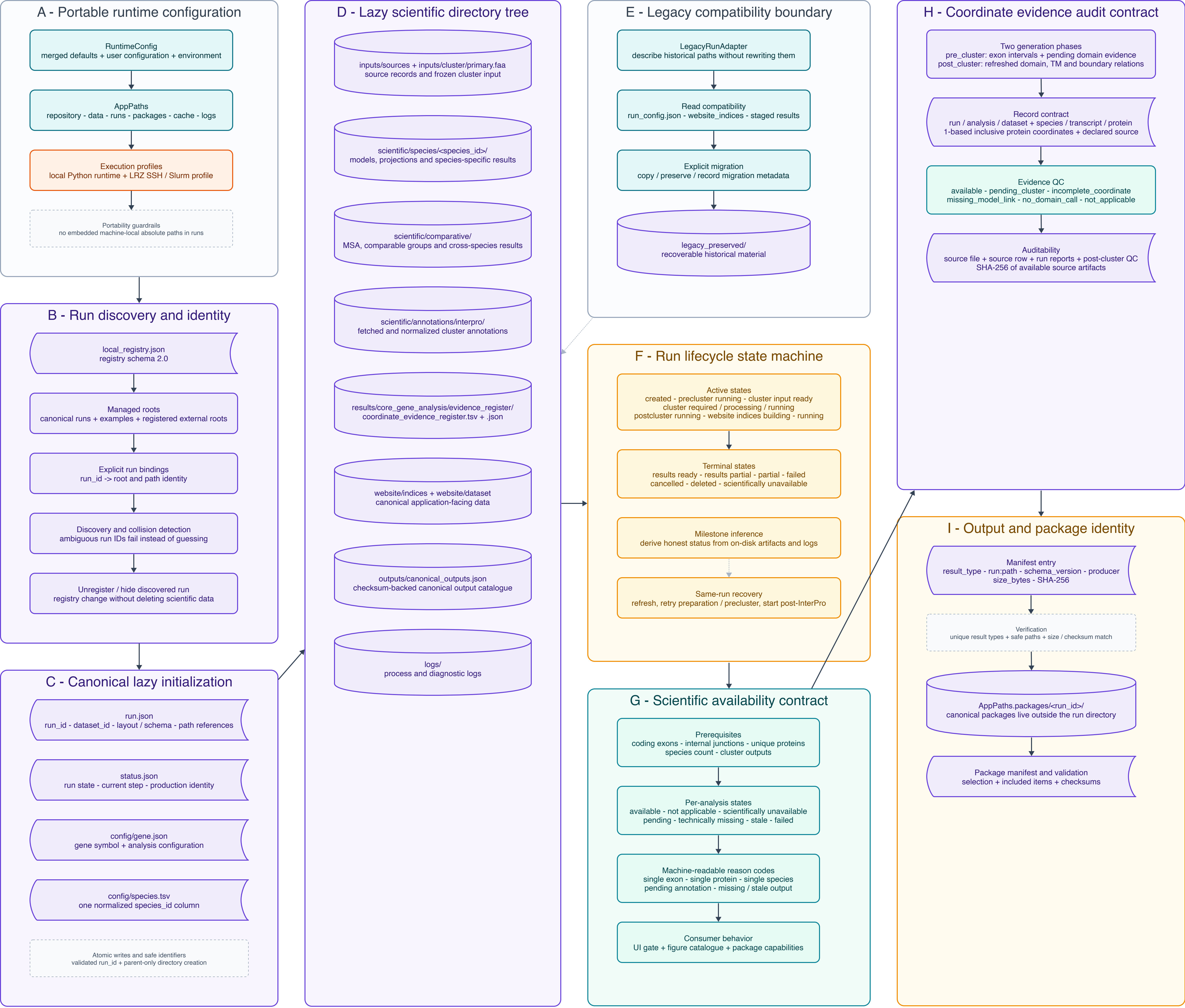


Detailed Run, Storage and State Contracts



State separation is deliberate: run lifecycle describes execution progress; analysis availability governs whether a result can be used; coordinate-record QC describes audit completeness. The evidence register is additive and run-local. It does not replace source tables or convert a missing domain call into a biological negative result.